COMS20018 — Mid-term Video Presentation (20%)

Overview

- **What:** an individual video presentation, up to **15 minutes** long.
- **Weight:** 20% of the unit mark.
- **Task:** choose **one question** from the list below and **answer it**. Your video is a discussion of that question, not a summary of a paper.
- **Scope:** Weeks 1–5.

Submission

- **Format:** a single **MP4** video file (.mp4). Other formats will not be accepted. Most recording tools (Zoom, Teams, PowerPoint, OBS, QuickTime) can export to MP4; check that your file plays before you submit it.
- **Where:** upload the file to the assignment on Blackboard.
- **Length:** up to **15 minutes**. Anything after 15:00 will not be watched or marked.
- **Deadline: Monday 2 Nov 2026 10:00am (GMT)**. Late submissions are subject to the University's late submission policy.

What we are looking for

1. **A clear answer to your question.** By the end of the video, a viewer should be able to say what your answer is and why you believe it.
2. **More than one view.** Look at the question from several angles. For example: theory vs. practice, one paper's claim vs. another's critique, one application vs. another, or what works vs. what can go wrong.
3. **Understanding grounded in the lectures.** Explain the concepts your answer depends on, accurately and in your own words.
4. **Good use of support documents.** Use sources to back up, challenge, or illustrate your answer. Don't just retell them.
5. **Creativity.** You decide how to present your answer. You could run a small experiment or demo (the lab notebooks are a good starting point), build an analogy or animation, stage a debate between two positions, or find a counterexample that breaks a common claim.

Support documents

Each question below comes with **example support documents**: papers, blog posts, videos and surveys. They are there to get you started. You do not have to use them.

- Use as many or as few support documents as your answer needs.

- You are **encouraged to find your own** support documents (papers, blogs, videos, textbook chapters, documentation, news articles), as long as every one is **properly cited**.
- **Citing:** show a reference list at the end of your video, and name the source on screen whenever you use its figure, result or idea. Give the author(s), title, year, and a venue or URL.
- Not all sources are equally reliable. A peer-reviewed paper, a blog post and a YouTube video carry different weight; say so when it matters to your argument.

Suggested structure

There is no required structure. This split is a guide to help you use the 15 minutes well:

Time	Content
~1–2 min	State your question, why it matters, and (briefly) your answer.
~3–4 min	Explain the concepts from the lectures that the answer depends on.
~6–7 min	Build your answer from several views, using your support documents and any demo or example of your own.
~2–3 min	Conclude: your answer, how confident you are, and what is still open.

Questions

Choose **one** of the questions below. Each lists the lectures it builds on, some possible angles (ideas, not a checklist), and example support documents.

Q1. What are overfitting and underfitting, and how can we avoid them?

Lecture link: Week 1 Lecture 2, Week 2

Possible angles: explain both through the bias–variance trade-off and the lecture's polynomial example; how to tell them apart from training vs. validation error, e.g. with cross-validation; ways to fix each.

Example support documents:

- Wilber & Werness. *Bias Variance Tradeoff.* MLU-Explain (2021). mlu-explain.github.io
- scikit-learn. *Underfitting vs. Overfitting* (example). scikit-learn.org
- Zhang, Bengio, Hardt, Recht, Vinyals. *Understanding Deep Learning Requires Rethinking Generalization.* ICLR 2017. arXiv:1611.03530

Q2. Pick a real-world AI application: which technique from the lectures does it use, and how does that technique make it work?

Lecture link: Week 1-5, refer to Week 1 Lecture 1 for applications

Possible angles: name the technique and explain how it works, as taught in the lectures; then show how it is used in the application: what goes in, what comes out, what data it is trained on, and what loss it minimises; why this technique suits this problem better than the alternatives we covered; where it falls short (e.g. a

model that scores well on its test set can still fail when deployed, or "human-level" claims that aren't a fair comparison). You may choose an application that is not in Week 1 Lecture 1.

Example support documents:

- Rajpurkar et al. *CheXNet: Radiologist-Level Pneumonia Detection on Chest X-Rays with Deep Learning.* arXiv:1711.05225 (2017).
- Norouzzadeh et al. *Automatically Identifying, Counting, and Describing Wild Animals in Camera-Trap Images with Deep Learning.* PNAS 115(25) (2018). doi:10.1073/pnas.1719367115
- Jumper et al. *Highly Accurate Protein Structure Prediction with AlphaFold.* Nature 596 (2021). doi:10.1038/s41586-021-03819-2

Q3. Describe the Transformer: how does it work, and why does it work so well?

Lecture link: Week 3 Lecture 1.

Possible angles: walk through self-attention, multi-head attention and positional encoding; what attention fixes compared with an RNN (long-range links, parallel training), and what it costs; encoder-only (BERT) vs. decoder-only (GPT) vs. both, by task; why a model built for text also works on images; is it the architecture, or the data and scale, that makes it work?

Example support documents:

- Vaswani et al. *Attention Is All You Need.* NeurIPS 2017. arXiv:1706.03762
- J. Alammar. *The Illustrated Transformer* (2018). jalammar.github.io
- 3Blue1Brown (G. Sanderson). *Attention in Transformers, Visually Explained* (2024). 3blue1brown.com

Q4. How do VAEs, GANs and diffusion models generate images, and what are their strengths and limitations?

Lecture link: Week 3 Lecture 2.

Possible angles: focus on two approaches and compare their training objectives and generation processes; discuss sample quality, diversity, speed and training difficulties, such as mode collapse; explain which approach might suit a chosen application and why.

Example support documents:

- Stanford CS236. *Deep Generative Models*, course notes. deepgenerativemodels.github.io
- Google for Developers. *Introduction to GANs.* developers.google.com
- Ho, Jain & Abbeel. *Denoising Diffusion Probabilistic Models.* NeurIPS 2020. arXiv:2006.11239

Q5. How does a GPT-style language model learn to predict the next token and use this to generate text?

Lecture link: Week 4 Lecture 1.

Possible angles: follow a simple example through tokenisation, embeddings, causal attention and the next-token probability distribution; explain how cross-entropy trains the model; compare greedy decoding and

sampling, including the effect of temperature; distinguish training from generation. A worked example or diagram could show how each generated token becomes context for the next prediction.

Example support documents:

- J. Alammar. *The Illustrated GPT-2* (2019). jalammar.github.io
- Jurafsky & Martin. *Speech and Language Processing*, Chapter 7: Transformers and Pretraining. web.stanford.edu
- Brown et al. *Language Models are Few-Shot Learners.* NeurIPS 2020. arXiv:2005.14165

Q6. Is Searle's argument in "Minds, brains and programs" correct?

Lecture link: Week 5 Lecture 1.

Possible angles: any of the critiques published alongside Searle's paper. To find them, log in to the University library, find the journal *Behavioral and Brain Sciences* and search for "Minds, brains, and programs". This can be tricky; if you get stuck, ask on Teams.

Example support documents:

- J. Searle. *Minds, Brains, and Programs.* Behavioral and Brain Sciences 3(3), 417–457 (1980). Pre-print
- Critiques of Searle, also in *Behavioral and Brain Sciences* 3(3). Available electronically via the University library, as above.

Q7. Do LLMs understand the answers they give us?

Lecture link: Week 5 Lecture 2.

Possible angles: why do hallucinations (sometimes) happen?; why are modern LLMs better (e.g. fewer hallucinations) than earlier models?; why does asking an LLM to explain itself help produce better answers?; what does it mean to "understand" something?

Example support documents:

- M. Wooldridge. *This Is Not the AI We Were Promised.* The Royal Society, lecture. YouTube

Marking criteria (out of 100)

Criterion	Weight	What we look for
Answer to the question	25%	A clear, direct answer to your chosen question, which the viewer can state after watching.
Understanding of the underlying concepts	25%	Accurate explanation of the lecture ideas your answer depends on, in your own words.
Multiple views & use of support documents	25%	The question is examined from more than one angle; sources are used to support, challenge or illustrate the answer, not just retold; all sources are cited.
Creativity & own insight	10%	Your own contribution: a demo, experiment, example, analogy, or argument that goes beyond the sources.
Communication, structure & format	15%	Clear, well organised, easy to follow, and within 15 minutes.

Band descriptors (applied holistically across the criteria above):

Band	Descriptor
70–100	Answers the question clearly and convincingly; explains the underlying concepts precisely and in your own words; weighs several views against each other and reaches a reasoned conclusion; uses well-cited sources critically; shows original thinking (e.g. own demo, example or argument); clear, well-paced, uses the time purposefully.
60–69	Answers the question with a clear position; mostly accurate explanation with minor gaps; considers more than one view, though not always in depth; sources used accurately and cited, with some critical comment; some own contribution; generally clear delivery.
50–59	Addresses the question but the answer is partial or implicit; correct but largely descriptive explanation; mostly one view; sources summarised rather than used to argue; little own contribution; adequate but unpolished delivery.
40–49	Weak link to the question; superficial or partly inaccurate explanation; sources referenced but not meaningfully used, or citations incomplete; disorganised delivery.
<40	Question not answered; explanation largely absent, incorrect, or irrelevant; no real use of sources, or sources used without citation.